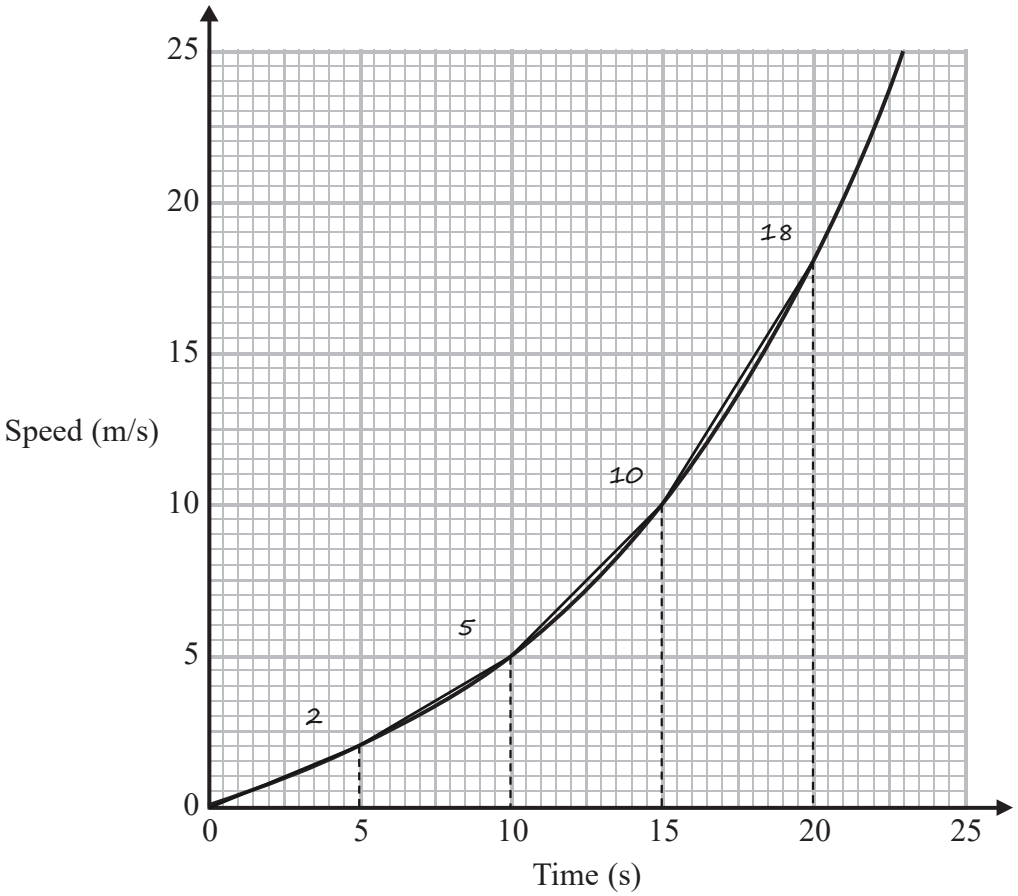
18 Here is a speed-time graph for a train.



(a) Work out an estimate for the distance the train travelled in the first 20 seconds.
Use 4 strips of equal width.

Each strip has width 20 ÷ 4 = 5 s. Trapezium area = ½(a + b)h.

$A_1 = ½(0 + 2) \times 5 = 5$

$A_2 = ½(2 + 5) \times 5 = 17.5$

$A_3 = ½(5 + 10) \times 5 = 37.5$

$A_4 = ½(10 + 18) \times 5 = 70$

1 mark
Correct area of a strip

1 mark
All four strips included

Distance ≈ 5 + 17.5 + 37.5 + 70 = 130 m

130 ..... m

1 mark
Correct distance

(3)

(b) Is your answer to (a) an underestimate or an overestimate of the actual distance the train travelled?
Give a reason for your answer.

Overestimate: the straight tops of the trapezia lie above the curve,

so they include extra area.

(1)

1 mark
Link the direction of error to the drawn trapezia

(Total for Question 18 is 4 marks)